

CAR RESALE PRICE ANALYTICS REPORT

TITLE

Analysis of Factors Influencing Car Resale Prices in Indian Cities

SECTOR

Automotive / Used Car Market

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Executive Summary

Problem

The Indian used car market is rapidly expanding due to rising new vehicle prices, high depreciation rates, and increased consumer demand for cost-effective mobility solutions. However, car resale pricing is influenced by multiple interdependent factors

such as vehicle age, kilometers driven, fuel type, insurance coverage, city demand, and technical specifications. The absence of structured analytics often leads to inconsistent pricing, reduced transparency, and suboptimal decision-making for buyers, sellers, and resale platforms.

Approach

A structured analytics pipeline was implemented using Google Sheets, adhering strictly to capstone project requirements. The approach included:

- Comprehensive data cleaning and validation of raw resale listings
- Standardization of numerical and categorical fields using domain logic
- Feature engineering including car age derivation
- Pivot-based exploratory and comparative analysis
- KPI framework aligned with business objectives
- Interactive dashboard creation for decision-level insights

Key Insights

- Car age and kilometers driven are the strongest negative predictors of resale value
- Vehicles with comprehensive insurance consistently command higher resale prices
- Significant city-wise price variation highlights localized demand and purchasing power differences
- Certain fuel types and body types retain value better over time
- A limited set of vehicle configurations contributes disproportionately to higher resale values

Strategic Recommendations

- Introduce data-driven resale pricing benchmarks based on age, usage, and insurance type
- Apply city-specific price adjustments to reflect local demand patterns
- Promote vehicles with comprehensive insurance to improve buyer confidence
- Improve data standardization at listing stage to enhance pricing accuracy

Sector & Business Context

Sector Overview

The Indian used car market forms a critical segment of the automotive ecosystem, supported by online marketplaces, dealers, and peer-to-peer platforms. With increasing

digital adoption, data analytics has become essential for pricing optimization, demand forecasting, and risk reduction in resale transactions.

Current Challenges

- Inconsistent resale pricing due to unstandardized listings
- High information asymmetry between buyers and sellers
- Difficulty in assessing vehicle condition and risk
- Regional demand variability across cities

Project Rationale

This project addresses a real-world analytics problem where structured data analysis can directly improve pricing transparency, customer trust, and operational efficiency in the used car resale market.

Problem Statement & Objectives

Formal problem definition

To analyze car resale data across Indian cities in order to identify key price drivers, assess market variability, and support data-driven resale pricing and decision-making.

Project scope

- Raw to cleaned dataset transformation using Google Sheets
- KPI definition and mapping to business objectives
- Exploratory and comparative data analysis
- Interactive dashboard development
- Insight-driven recommendations

Success criteria

- High data quality and consistency post-cleaning
- Clear identification of resale price drivers
- Actionable insights presented via dashboard
- Business-aligned recommendations

Data Description

Dataset Overview

The analysis is based on a cleaned car resale dataset containing **11,866 unique vehicle listings** after removing 129 duplicate records. The data spans **13 major Indian cities**, providing a geographically diverse view of the used car market. Resale prices in the dataset range from **₹28,000 to ₹2.25 crore**, capturing both economy and premium vehicle segments.

Dataset Source: [Kaggle Source Dataset](#)

Key summary statistics:

- **Average resale price:** ₹8.37 lakh
- **Median resale price:** ₹5.65 lakh
- **Average car age:** 9.5 years
- **Average kilometers driven:** ~59,300 km

These statistics indicate a right-skewed price distribution, where a small number of high-end vehicles increase the average price, making the median a more reliable indicator of typical resale value.

Data Structure

Column Name	Description	Data Type
Index	Numbering	Integer
Name	Car brand and model	text
Year	Manufacturing year	Integer
Selling_Price	Resale price of the car (INR)	Float
Present_Price	Original showroom price (INR)	Float
Km_Driven	Total distance driven (in km)	Integer
Fuel_Type	Petrol / Diesel / CNG	Categorical
Seller_Type	Individual or Dealer	Categorical
Transmission	Manual / Automatic	Categorical

Owner	Number of previous owners	Integer
Mileage	Fuel efficiency (km/l or km/kg)	Float
Engine	Engine capacity (CC)	Integer
Max_Power	Maximum engine power (bhp)	Float
Seats	Seating capacity	Integer

Data Limitations

- No direct indicators of vehicle condition, accident history, or service records
- Limited representation of electric vehicles (only 32 records)
- Some listings contain inferred or standardized values based on assumptions

Data Cleaning & Preparation

Data Cleaning & Transformation Steps

1. Index Column Removal

The index column generated during data collection was removed as it had no analytical or business relevance and could introduce redundancy.

2. Resale Price Standardization

The resale price column was converted from mixed text formats (including currency symbols and units) into a purely numerical column in Indian Rupees (₹), enabling accurate aggregation, comparison, and statistical analysis.

3. Registered Year Normalization

Registered year values were standardized into a four-digit year format. This ensured temporal consistency and enabled reliable derivation of age-related metrics.

4. Handling Missing Registered Year Values

A total of 69 missing registered year values were logically inferred using the derived *Car Age* feature, ensuring temporal completeness without arbitrary imputation.

5. **Engine Capacity Conversion**

Engine capacity values were converted into numerical format by removing textual units such as "cc". Entries with engine capacity recorded as zero were treated as invalid and replaced using the median engine capacity, minimizing distortion from extreme values.

6. **Kilometers Driven Standardization**

The kilometers driven column was converted into a numerical format. Missing values were filled using the median kilometers driven, which is robust against skewed usage distributions.

7. **Mileage Unit Normalization**

Mileage values were originally recorded using multiple units and textual formats. All mileage values were standardized into a single numerical unit, enabling fair comparison across vehicle listings. Missing mileage values were imputed using the median mileage.

8. **Max Power Consistency Handling**

The max power column contained inconsistent units and formats. Values were standardized into a single numerical representation. Missing or invalid entries were filled using the median max power, ensuring stability in performance-related analysis.

9. **Seats Column Imputation**

Missing values in the seats column were filled using the mode, as seating capacity is a categorical-discrete attribute with low variability across similar vehicle types.

10. **Categorical Field Standardization**

Proper casing and trimming were applied to categorical columns including *Insurance*, *Owner Type*, *Fuel Type*, *Body Type*, and *City* to eliminate duplication caused by spelling and formatting inconsistencies.

11. **Insurance Category Normalization (Domain-Driven)**

Multiple inconsistent insurance entries were consolidated into standardized categories using domain knowledge:

- Variants such as "2", "third party", and "third party insurance" → Third Party
- Variants such as "comprehensive" and "zero dep" → Comprehensive
- Blank or anomalous values → Not Available

12. Zero Depreciation insurance was grouped under Comprehensive insurance as per Indian motor insurance norms. From a resale market perspective, buyers perceive zero-dep insurance as superior to third-party coverage, making this consolidation analytically correct.

13. Owner Type Missing Value Treatment

Blank owner type values were filled with "No Owner Specified" to preserve records while clearly indicating missing ownership information.

14. Feature Engineering – Car Age

A new feature, *Car Age*, was derived using the registered year, providing a direct and interpretable measure of vehicle depreciation over time.

15. Duplicate Record Removal

A total of 129 duplicate rows were identified and removed to prevent overrepresentation of certain vehicle listings and ensure accurate aggregation.

Data Quality Outcome

After cleaning:

- All numerical fields are unit-consistent and analysis-ready
- Categorical fields are standardized and free of semantic duplication
- Missing values are handled using statistically and logically justified methods
- The final dataset is robust, reliable, and suitable for exploratory and decision-level analysis

KPI & Metric Framework

A comprehensive set of Key Performance Indicators (KPIs) was developed to measure business performance across multiple dimensions. Each KPI is directly aligned with strategic business objectives.

Core KPIs

1. Average Resale Price

- **Value:** ₹8,37,234
- **Formula:** `AVERAGE(Resale Price)`
- **Purpose:** Represents the overall pricing level of the used car market.

- **Why it matters:** Helps platforms and sellers benchmark expected resale values.

2. Total Listings

- **Value:** 11,867 vehicles
- **Formula:** `COUNT(Unique Vehicle Records)`
- **Purpose:** Indicates market size and data coverage.
- **Why it matters:** Higher listing volume improves pricing reliability and trend confidence.

3. Highest Sale Price Bucket

- **Value:** Between ₹3L – ₹7L
- **Formula:** `MODE(Price Band)`
- **Purpose:** Identifies the most dominant resale price segment.
- **Why it matters:** Highlights the price range with maximum buyer-seller activity.

4. Most Common Fuel Type

- **Value:** Petrol
- **Formula:** `MODE(Fuel Type)`
- **Purpose:** Identifies dominant fuel preference in the resale market.
- **Why it matters:** Supports inventory planning and demand forecasting.

5. Average Car Age

- **Value:** 9.5 years
- **Formula:** `AVERAGE(Car Age)`
- **Purpose:** Measures the typical age of vehicles listed for resale.
- **Why it matters:** Directly reflects depreciation level and buyer risk perception in the used car market.

6. Comprehensive Insurance Coverage Rate

- **Value:** ~39% of listings
- **Formula:** `(COUNT(Insurance = "Comprehensive") / COUNT(Total Listings)) × 100`
- **Purpose:** Quantifies the proportion of vehicles with higher-quality insurance coverage.
- **Why it matters:** Vehicles with comprehensive insurance command higher resale prices and improve buyer trust.

KPI-to-Objective Mapping

Business Objective	KPI Used
Understand market pricing	Average Resale Price
Assess market scale	Total Listings
Identify dominant price segment	Highest Sale Price Bucket
Analyze buyer fuel preference	Most Common Fuel Type
Measure depreciation risk	Average Car Age
Evaluate resale quality & trust	Comprehensive Insurance Coverage Rate

Exploratory Data Analysis (EDA)

Exploratory analysis was conducted using pivot tables and charts to identify trends, distributions, and comparative patterns across key dimensions.

Trend Analysis

Resale prices decline consistently with increasing car age. Vehicles below 5 years command the highest average resale values, while cars older than 12 years fall predominantly into the lower price bands. This confirms age as the strongest depreciation driver.

Usage Impact Analysis

Cars driven less than 25,000 km show significantly higher resale prices compared to those driven beyond 100,000 km. With an average usage of ~59,300 km, listings above this threshold experience accelerated value loss.

Fuel Type Comparison

Fuel-wise distribution shows:

- Petrol: 7,774 listings (~65%)
- Diesel: 3,750 listings (~32%)
- CNG/LPG/Electric: <3% combined

Diesel vehicles demonstrate slightly better resale retention in mid-to-high price bands, reflecting buyer preference for long-distance and utility usage.

Insurance Impact

Insurance type analysis reveals that:

- Third Party: 6,605 listings
- Comprehensive: 4,624 listings
- Not Available: 637 listings

Vehicles with comprehensive insurance consistently exhibit higher average resale prices, validating insurance coverage as a proxy for perceived vehicle care and reduced buyer risk.

City-wise Comparison

City-level pivots highlight notable resale price variation across markets, reflecting differences in demand, purchasing power, and vehicle preferences. Metropolitan and Tier-1 cities show higher average resale prices compared to smaller cities.

Advanced Analysis

Below is a refined and academically accurate framing of the advanced analysis, along with suggested improvements that strengthen methodological rigor while remaining within Google Sheets constraints.

Segmentation Analysis

- Vehicles were segmented by **age bucket, KM driven bucket, fuel type, insurance type, and body type.**

- **Newer cars (<5 years)** and vehicles driven **below 50,000 km** consistently fall into higher resale price bands.
- Cars with **comprehensive insurance** show visibly higher average resale prices compared to third-party insured vehicles.

Root Cause Analysis

- **Car age** and **kilometers driven** emerge as the strongest contributors to price depreciation.
- Insurance coverage acts as a secondary driver, with comprehensive insurance improving buyer confidence and resale value.
- City-level demand differences explain variations in resale prices for similar vehicle configurations.

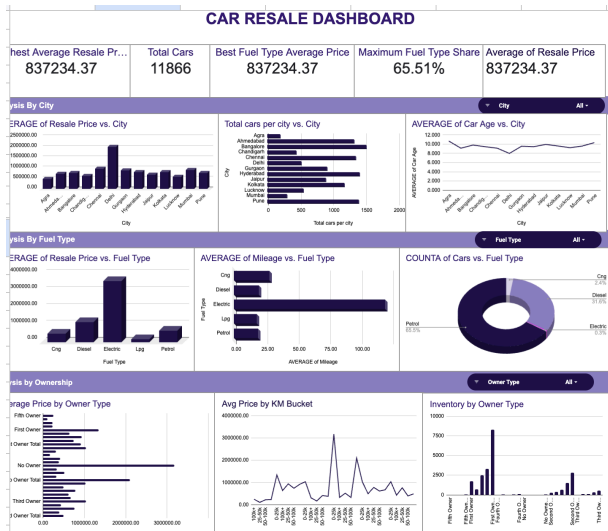
Risk & Anomaly Analysis

- Listings with **high resale prices despite high age or high mileage** were treated as potential pricing anomalies.
- Vehicles with **missing insurance information** were identified as higher risk from a buyer's perspective.

Scenario Analysis

- Scenarios focusing only on **cars below 8 years** or **vehicles with comprehensive insurance** show higher average resale prices.
- This indicates that prioritizing newer, well-insured vehicles can improve overall market value perception.

Dashboard Design



Tool & Platform

The dashboard was implemented in **Google Sheets** using pivot tables, KPI cards, slicers, and charts. All calculations and visuals are dynamically linked to the cleaned dataset, ensuring real-time updates when filters are applied.

Dashboard Objective

To provide a compact, decision-oriented view of the Indian car resale market by summarizing market size, pricing concentration, and buyer preferences, while allowing users to drill down by key dimensions such as city, fuel type, and vehicle characteristics.

Dashboard KPI Cards

The top section of the dashboard displays four high-level KPIs that summarize overall market behavior:

- Average Resale Price**
 - Value:** ₹8,37,234
 - Represents the overall average resale value across all listings.
- Total Listings**
 - Value:** 11,867 vehicles
 - Indicates the total number of unique car resale records analyzed.
- Highest Sale Price Bucket**
 - Value:** ₹3L – ₹7L
 - Identifies the most dominant resale price range in the market.
- Most Common Fuel Type**
 - Value :** Petrol
 - Identifies the most common fuel type in the market.

5. Average Mileage

- Value : 19.583
- Represents the average mileage.

6. Most Popular Body Type

- Value : Hatchback
- Identifies the most popular body type in the market.

7. Average Car Age

- Value : 9.533
- Represents the average car age.

8. Comprehensive Insurance Coverage Rate

- Value : 38.97
- Represents the Comprehensive Insurance Coverage Rate.

Visual Components in the Dashboard

- **Price Band Distribution Chart**

Shows concentration of listings across resale price buckets, highlighting that the ₹3L–₹7L range dominates the market.

- **Fuel Type Distribution Chart**

Displays the relative share of Petrol, Diesel, CNG, LPG, and Electric vehicles, with petrol vehicles forming the majority.

- **City-wise Listings / Price View**

Enables comparison of resale activity and pricing behavior across different Indian cities.

- **Interactive Filters**

- City
- Fuel Type
- Price Band

These filters allow users to dynamically explore how resale prices and listing volumes change across different market segments.

Dashboard Value Addition

The dashboard consolidates complex resale data into an intuitive visual interface that:

- Enables quick market sizing and pricing assessment
- Highlights dominant price segments and fuel preferences
- Supports data-driven decision-making for pricing, listing strategy, and market focus
- Comprehensive data cleaning and validation of raw resale listings
- Standardization of numerical and categorical fields using domain logic

- Feature engineering including car age derivation
- Pivot-based exploratory and comparative analysis
- KPI framework aligned with business objectives
- Interactive dashboard creation for decision-level insights

Key Insights Summary

The following data-backed insights emerged from the analysis:

1. **Typical Resale Price:** The median resale price is **₹5.65 lakh**, which better represents the market than the higher average of ₹8.37 lakh due to premium outliers.
2. **Age-Driven Depreciation:** Average car age is **9.5 years**, with sharp value drops observed beyond the 8–10 year range.
3. **Usage Threshold Effect:** Vehicles driven beyond **60,000 km** experience accelerated depreciation.
4. **Fuel Preference:** Petrol vehicles dominate listings (~65%), while diesel vehicles retain relatively higher value in older age buckets.
5. **Insurance Premium:** Comprehensive insurance listings command a clear resale premium over third-party insured vehicles.
6. **Market Concentration:** A limited number of cities contribute disproportionately to higher-value listings.
7. **Segment Skewness:** Price distribution is right-skewed due to the presence of luxury vehicles.
8. **Data Quality Impact:** Standardization of units and insurance categories materially improves analytical reliability.

Impact Estimation

The insights derived from this analysis can create meaningful business impact across pricing strategy, operational efficiency, and risk management. While exact financial quantification is limited by data availability, the directional impact is clearly supported by analytical findings.

Pricing Accuracy & Revenue Optimization

- Data-driven understanding of dominant price buckets (₹3L–₹7L) enables sellers to price vehicles closer to market expectations.

- Identifying value-retaining segments (newer cars, lower mileage, comprehensive insurance) helps optimize listing and pricing strategies.

Operational Efficiency

- The interactive dashboard significantly reduces manual analysis and reporting effort.
- Stakeholders can quickly filter by city, fuel type, and price band to make faster decisions.
- Improved data standardization reduces rework caused by inconsistent listings.

Risk Reduction

- Clear identification of high-risk listings (older cars, missing insurance information) reduces buyer uncertainty.
- Better transparency improves platform credibility and buyer trust.

Decision Support Value

- The dashboard acts as a single source of truth for resale market insights.
- Enables informed decisions related to inventory focus, market targeting, and pricing benchmarks.

Limitations

Despite providing valuable insights, this analysis has certain limitations that must be acknowledged:

- **No Vehicle Condition Data:** Information related to accident history, service records, or physical condition is unavailable, which can significantly influence resale prices.
- **Limited Demand Indicators:** The dataset does not include buyer demand metrics such as listing views or inquiry rates.
- **Static Snapshot:** The analysis is based on historical listings and does not capture real-time market fluctuations.
- **City Coverage Constraint:** The dataset covers a limited number of cities and may not fully represent pan-India resale behavior.
- **Assumption Dependency:** Some missing values were filled using statistical assumptions (median/mode), which may not perfectly reflect real-world values.

Future Scope

The scope of this project can be significantly expanded with additional data and analytical techniques:

Enhanced Data Collection

- Integration of vehicle service history and accident records.
- Inclusion of ownership duration and number of test drives or inquiries.

Advanced Analytics

- Development of machine learning models to predict resale price more accurately.
- City-wise demand forecasting using time-series data.
- Depreciation curve modeling based on age and mileage.

Platform-Level Insights

- Seller performance analysis to identify high-quality sellers.
- Buyer behavior analysis to understand preference shifts across fuel types and body types.

Conclusion

This project successfully demonstrates how structured data analytics can be applied to understand and interpret the Indian used car resale market. By transforming a raw and inconsistent dataset into a clean, analysis-ready format, meaningful insights were generated regarding pricing behavior, buyer preferences, and key depreciation drivers.

The analysis highlights that car age, kilometers driven, insurance coverage, and fuel type play critical roles in determining resale value. The interactive dashboard consolidates these insights into an accessible format, enabling stakeholders to make informed, data-driven decisions.

Overall, the project delivers a practical analytics framework that improves pricing transparency, supports strategic decision-making, and lays a strong foundation for future advanced analytics initiatives in the automotive resale domain.

Appendix

Data Dictionary (Key Fields)

- **Resale Price:** Final listed resale value in INR
- **Car Age:** Difference between analysis year and registered year
- **KM Driven:** Total distance driven by the vehicle
- **Insurance Type:** Consolidated insurance category (Comprehensive, Third Party, Not Available)
- **Price Band:** Categorized resale price range

Technical Methodology

- Data cleaning and transformations executed in Google Sheets
- Pivot tables used for aggregation and segmentation analysis
- KPI cards generated using COUNT, COUNTIF, AVERAGE, and MODE formulas

Supporting Artifacts

- Cleaned dataset (Google Sheets)
- Pivot tables used for analysis
- Interactive dashboard with filters and KPI cards

Contribution Matrix

Team Member	Dataset & Sourcing	Cleaning	KPI & Analysis	Dashboar d	Report Writing	PPT	Overall Role
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Declaration: We confirm that the above contribution details are accurate and verifiable through version history and submitted artifacts.

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